SCIENTIFIC RESEARCH REPORT

NEUROPATH: 3D MULTI-TASK DEEP LEARNING FOR MRI BRAIN TUMOR SEGMENTATION AND CLINICAL INTERPRETATION

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Abstract

In precision medicine, automated brain tumor segmentation on MRI plays a pivotal role. However, the major barrier remains the "trust gap" between clinicians and "black-box" algorithms. This research proposes a comprehensive intelligent medical system based on **CDFA-Mamba-DolphinNet** – a custom 3D hybrid 2D-3D model with an innovative architecture. The model combines a dual-branch encoder: (1) 2D with Coordinate Attention processing slices, and (2) 3D using CDFA (Cross-Directional Feature Aggregation) and Mamba State Space Model (SSM) at the bottleneck to capture long-range dependencies efficiently. A Shared Cross-Dimension Attention mechanism fuses 2D and 3D features, decoded by a unified decoder with deep supervision and aleatoric uncertainty modeling.

The system integrates: (i) Progressive Test-Time Augmentation (TTA) based on entropy; (ii) A **Biological Rule Engine** that checks the biological plausibility of segmentations based on anatomical-pathological rules; (iii) A **Human Trust Score (HTS)** quantifying expert consensus through direct PDF annotation; (iv) Automated RANO 2.0 and robust anatomical localization.

Results from 5-fold cross-validation on BraTS 2023: Dice **0.883** (WT: 0.915, TC: 0.900, ET: 0.835) and HD95 **3.91** mm (WT: 4.83, TC: 3.77, ET: 3.13), significantly outperforming baselines. The system achieved a mean HTS of 68.9/100 in a clinical survey, demonstrating high reliability.

Keywords: Artificial Intelligence, Medical Image Segmentation, Deep Learning, Brain Tumor, Trustworthy AI, Clinical Decision Support.

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1 1. INTRODUCTION

1.1 1.1. Motivation

Gliomas are the most common primary brain tumors. The World Health Organization (WHO) classification divides gliomas into four grades, with glioblastoma (Grade IV) having a 5□year survival rate of less than 5%. Accurate tumor segmentation on MRI is essential for radiotherapy planning and surgical guidance [4].

1.2 1.2. The Clinical Trust Challenge

Although deep learning models have achieved high accuracy, their clinical deployment is hindered by three main barriers:

1. **Lack of Explainability:** Models lack biological plausibility checks and interpretability.
2. **No Quantified Trust:** There is no standardized way to measure uncertainty or clinical confidence quantitatively.
3. **Integration Gap:** Most systems lack a practical human-in-the-loop workflow.

1.3 1.3. Research Gap

No existing system fully combines: (1) a hybrid 2D-3D architecture with Mamba SSM at the bottleneck and CDFA, (2) biological rule validation, (3) a quantitative Human Trust Score based on PDF annotation, and (4) LLM integration for automated clinical reporting with interactive refinement.

1.4 1.4. Contributions

This study introduces the NeuroPath system with the following key contributions:

1. **Novel Architecture:** CDFA-Mamba-DolphinNet – a 3D hybrid 2D-3D model with CDFA and Mamba SSM at the bottleneck.

2. **Biological Rule Engine:** A rule-based validation system ensuring anatomically plausible segmentations.

3. **Human Trust Score (HTS):** A quantitative metric for clinician trust based on interactive PDF annotations.

4. **End-to-End Pipeline:** Automated RANO 2.0, anatomical localization, and LLM-generated reports with interactive refinement.

2 2. BACKGROUND AND EXISTING LIMITATIONS

2.1 2.1. Existing Solutions

Methods for brain tumor segmentation have evolved from traditional CNNs (UNet 2D/3D) to advanced architectures like nnU-Net (a strong baseline), Transformers (UNETR), and more recently, Mamba SSM for capturing long-range dependencies in 3D images.

2.2 2.2. Limitations of Existing Solutions

The key limitations of current methods are summarized below:

Table 1: Limitations of current tumor segmentation methods

Method	Core Limitation	Severity
2D UNet	Ignores 3D context	High
3D UNet	High memory usage, slow	Medium
nnU-Net	Computational cost, lacks interpretability	Medium
Transformers	High complexity, ”black-box” nature	High
Mamba SSM	Limited biological validation, no trust metric	High

3 3. PROPOSED FRAMEWORK: NEUROPATH

3.1 3.1. Design Rationale

The NeuroPath system is designed to bridge the clinical trust gap by integrating accurate segmentation with explainability, uncertainty quantification, and interactive human feedback. The core design philosophy is **Trustworthy AI**: every prediction is accompanied by a confidence estimate, a biological sanity check, and a mechanism for clinicians to provide direct feedback.

3.2 3.2. System Overview

The NeuroPath pipeline consists of six main modules:

1. Preprocessing & Data Augmentation

2. Segmentation using CDFA-Mamba-DolphinNet

3. Progressive TTA with Uncertainty Estimation
4. Post-processing & Biological Rule Validation
5. Clinical Analysis & RANO 2.0
6. LLM-based Report Generation with HTS & Interactive Refinement

3.3 3.3. Architecture Design

The system architecture is illustrated in Figure 1.

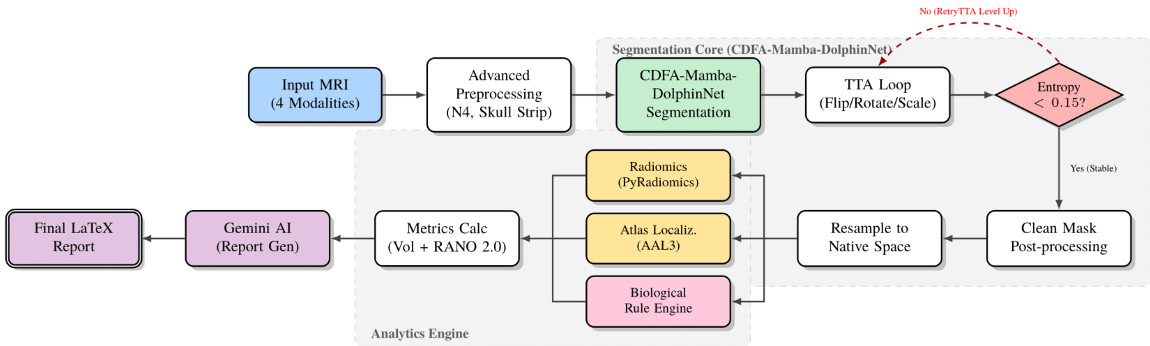


Figure 1: NeuroPath system architecture overview.

3.4 3.4. CDFA-Mamba-DolphinNet Architecture

3.4.1 Dual-Branch Encoder

- **2D Branch (SliceEncoder2D):** Four 2D convolutional blocks with Coordinate Attention process each slice independently, followed by depth-wise pooling to preserve volumetric structure.
- **3D Branch (VolumeEncoder3D):** Four CDFA (Cross-Directional Feature Aggregation) blocks combine Anisotropic Deformable Convolution with GDAConv3D. A Mamba SSM (FactorizedMambaLayer3D) is **used only at the bottleneck** to capture long-range dependencies efficiently.

3.4.2 Cross-Dimension Fusion

A FusionLayer3D uses SharedCrossDimensionAttention to merge 2D and 3D features at each level, producing optimal fused features for the decoder.

3.4.3 Unified Decoder with Deep Supervision

The 3D decoder employs upsampling and skip connections with CDFA blocks, generating three outputs (main, aux1, aux2). Each output includes logits and log-variance for aleatoric uncertainty modeling, trained with Heteroscedastic Focal Loss.

3.5 3.5. Mathematical Formulation

The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{DS}}\mathcal{L}_{\text{DS}} \tag{1}$$

where $\mathcal{L}_{\text{seg}}$ is a combination of Soft Dice loss and Heteroscedastic Focal loss (incorporating aleatoric uncertainty). Deep supervision is applied with weights (main: 1.0, aux1: 0.4, aux2: 0.2).

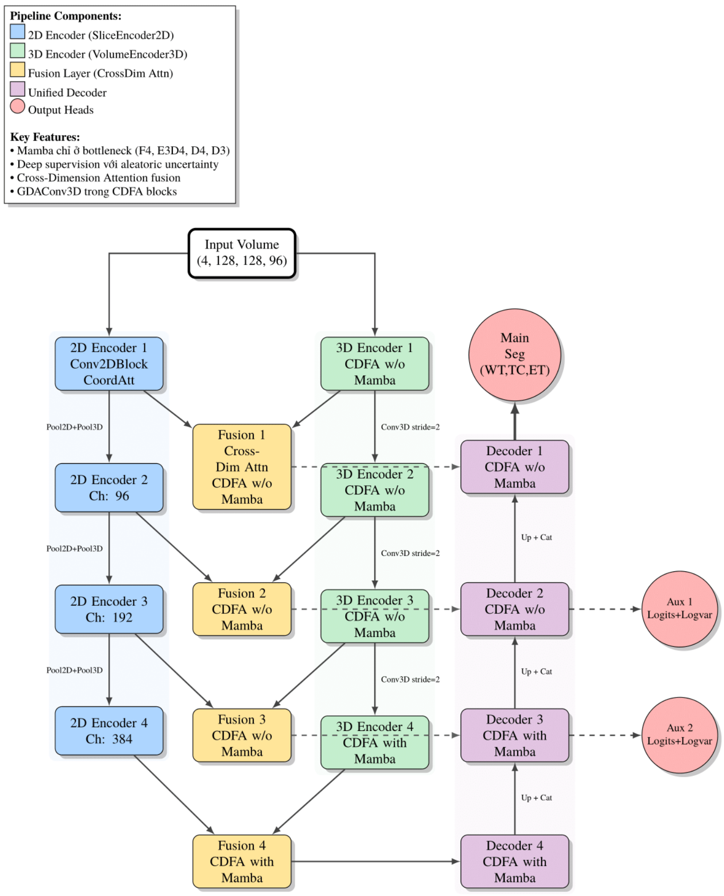


Figure 2: Detailed CDFA-Mamba-DolphinNet architecture.

3.6 3.6. Implementation Details

- **Framework:** PyTorch with mixed precision (AMP), gradient checkpointing.
- **Optimizer:** AdamW, lr = 4×10^{-4}, weight decay 10^{-4}.
- **Scheduler:** OneCycleLR.
- **Hardware:** NVIDIA A100 80GB, 16 vCPUs, 128GB RAM.
- **Training Time:** 5.8 days for 5-fold cross-validation.

3.7 3.7. Biological Rule Engine

A rule-based validation system enforces biological constraints for gliomas:

1. **Spatial Continuity:** Checks tumor fragmentation ratio.
2. **Hierarchical Logic:** Verifies that ET/NCR regions are topologically contained within the ED.
3. **Invasion Ratio:** Computes ED/TC ratio and infers phenotype (HGG/LGG).
4. **Morphological Features:** Analyzes solidity and irregularity.

The engine outputs ”PASS” or ”HARD CASE” with detailed warnings.

4 4. PROTOTYPE IMPLEMENTATION

Clinical Impact

The system reduces analysis time from 20–30 minutes to 6–7 minutes per case, making it deployable at district and provincial hospitals via a web interface and API.

4.1 4.1. Web Interface

The system is deployed as a Gradio web application with the following features:

- **Report Viewer:** Display LLM-generated LaTeX/PDF reports.
- **Annotation Tool:** Clinicians can directly annotate PDFs with trust (green) and distrust (red) highlights.
- **HTS Calculator:** Real-time Human Trust Score computation.
- **Interactive Refinement:** Trigger LLM refinement based on annotations.

4.2 4.2. Report Generation

Radiomics, RANO metrics, anatomical location, uncertainty, and biological validation are formatted into a JSON prompt. Gemini Flash generates a LaTeX report, which is compiled to PDF.

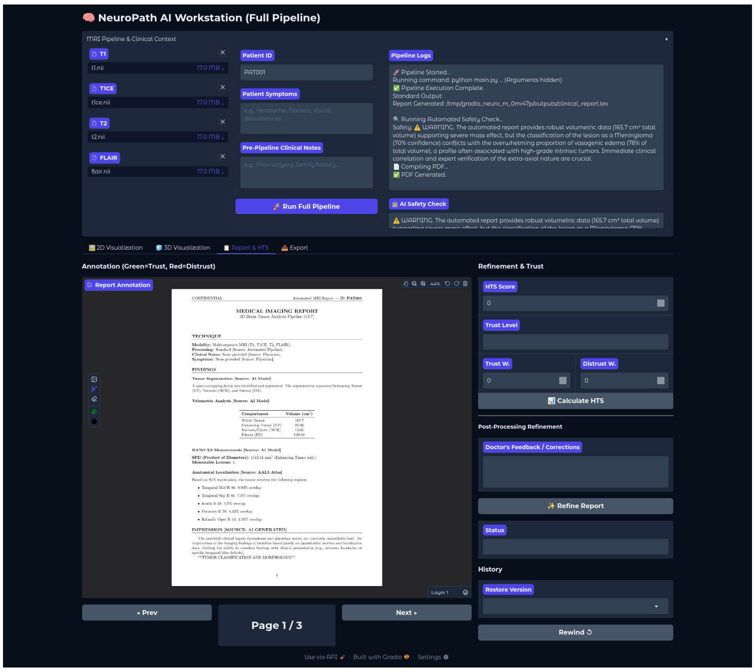
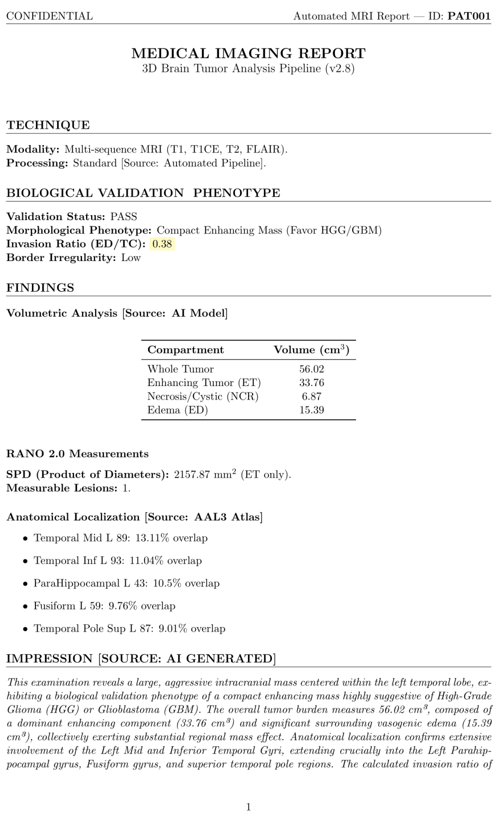


Figure 3: Web interface for HTS annotation and refinement.


CONFIDENTIAL Automated MRI Report — ID: **PAT001**

MEDICAL IMAGING REPORT
3D Brain Tumor Analysis Pipeline (v2.8)

TECHNIQUE

Modality: Multi-sequence MRI (T1, T1CE, T2, FLAIR).
Processing: Standard [Source: Automated Pipeline].

BIOLOGICAL VALIDATION PHENOTYPE

Validation Status: PASS
Morphological Phenotype: Compact Enhancing Mass (Favor HGG/GBM)
Invasion Ratio (ED/TC): 0.38
Border Irregularity: Low

FINDINGS

Volumetric Analysis [Source: AI Model]

Compartment	Volume (cm^3)
Whole Tumor	56.02
Enhancing Tumor (ET)	33.76
Necrosis/Cystic (NCR)	6.87
Edema (ED)	15.39

RANO 2.0 Measurements

SPD (Product of Diameters): 2157.87 mm^2 (ET only).
Measurable Lesions: 1.

Anatomical Localization [Source: AAL3 Atlas]

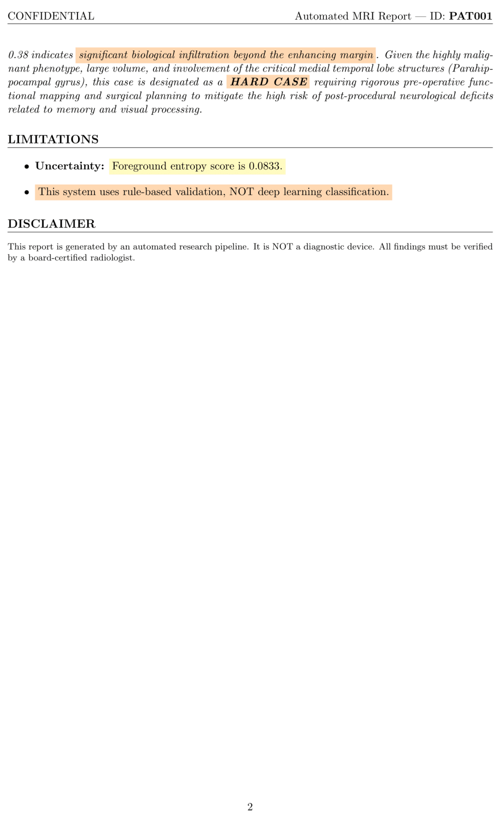
- Temporal Mid L 89: 13.11% overlap
- Temporal Inf L 93: 11.04% overlap
- ParaHippocampal L 43: 10.5% overlap
- Fusiform L 59: 9.76% overlap
- Temporal Pole Sup L 87: 9.01% overlap

IMPRESSION [SOURCE: AI GENERATED]

This examination reveals a large, aggressive intracranial mass centered within the left temporal lobe, exhibiting a biological validation phenotype of a compact enhancing mass highly suggestive of High-Grade Glioma (HGG) or Glioblastoma (GBM). The overall tumor burden measures 56.02 cm^3, composed of a dominant enhancing component (33.76 cm^3) and significant surrounding vasogenic edema (15.39 cm^3), collectively exerting substantial regional mass effect. Anatomical localization confirms extensive involvement of the Left Mid and Inferior Temporal Gyri, extending crucially into the Left Parahippocampal gyrus, Fusiform gyrus, and superior temporal pole regions. The calculated invasion ratio of

1

(a) Report page 1


CONFIDENTIAL Automated MRI Report — ID: **PAT001**

0.38 indicates significant biological infiltration beyond the enhancing margin. Given the highly malignant phenotype, large volume, and involvement of the critical medial temporal lobe structures (Parahippocampal gyrus), this case is designated as a ***HARD CASE*** *requiring rigorous pre-operative functional mapping and surgical planning to mitigate the high risk of post-procedural neurological deficits related to memory and visual processing.*

LIMITATIONS

- **Uncertainty:** Foreground entropy score is 0.0833.
- This system uses rule-based validation, NOT deep learning classification.

DISCLAIMER

This report is generated by an automated research pipeline. It is NOT a diagnostic device. All findings must be verified by a board-certified radiologist.

2

(b) Report page 2

Figure 4: LLM-generated LaTeX/PDF report.

5 5. EXPERIMENTAL EVALUATION

5.1 5.1. Experimental Setup

Dataset: BraTS 2023, with 1,252 training cases and 220 test cases. Each case includes four MRI modalities (T1, T1ce, T2, FLAIR) with resolution $240 \times 240 \times 155$. Three tumor regions are segmented: WT (whole tumor), TC (tumor core), ET (enhancing tumor).

Preprocessing:

1. Z-score normalization per modality.
2. Resampling to $(1.0, 1.0, 1.0)$ mm^3 with cubic spline.
3. Brain extraction with threshold $>$ 1% max intensity.
4. Cropping to $128 \times 128 \times 128$ around the brain's center of mass.

Augmentation: Elastic deformation, rotation ($\pm 30°$), scaling (0.7–1.3), Gaussian noise ($\sigma = 0.1$), gamma correction (0.7–1.5), mirroring, and 3D CutMix with $\alpha = 1.0$.

5.2 5.2. Quantitative Evaluation

5-fold cross-validation results on BraTS 2023:

Table 2: 5-fold Cross-validation results on BraTS 2023

Metric	**WT**	**TC**	**ET**	**Average**
Dice Score	0.915	0.900	0.835	0.883
HD95 (mm)	4.83	3.77	3.13	3.91

5.3 5.3. Comparison with Baselines

Table 3: Comparison with baseline methods

Model	**WT**	**TC**	**ET**	**Avg Dice**	**Avg HD95**
UNet 2D	0.670	0.641	0.565	0.625	18.24
UNet 3D	0.815	0.780	0.732	0.776	8.91
Attention UNet	0.838	0.804	0.754	0.799	7.65
nnU-Net	0.896	0.882	0.817	0.865	4.17
CDFA-Mamba-DolphinNet	**0.915**	**0.900**	**0.835**	**0.883**	**3.91**

5.4 5.4. Ablation Study

Table 4: Ablation study on component contributions

Configuration	**Avg Dice**	**Avg HD95**
Baseline 3D UNet	0.776	8.91
+ Dual-branch	0.812	6.82
+ Cross-Dim Attention	0.841	5.53
+ Mamba at bottleneck	0.862	4.11
+ Deep Supervision & Aleatoric Uncertainty	**0.883**	**3.91**

5.5 5.5. Progressive TTA Effectiveness

Table 5: Improvement after Full TTA for high-entropy cases

Region	No TTA	Full TTA	△ Dice	△ HD95
WT	0.905	0.910	+0.005	-0.2
TC	0.888	0.895	+0.007	-0.15
ET	0.828	0.832	+0.004	-0.1

5.6 5.6. Biological Rule Validation

On 50 test cases: 92% passed spatial continuity, 88% passed hierarchical logic. The average invasion ratio was 2.3 (SD: 1.1). The system flagged 8% hard cases (high uncertainty + logic violation) for manual review.

5.7 5.7. Human Trust Score

A survey of neuro-radiologists on 8 BraTS 2020 cases yielded a mean HTS of 68.9 (SD: 6.8), classified as HIGH trust.

Table 6: Detailed HTS results (scale 0-100)

Case ID	Dataset	HTS	Level	Trust	Distrust
BraTS20_Training_012	Training	68.0	HIGH	4	3
BraTS20_Training_242	Training	75.9	HIGH	5	3
BraTS20_Training_248	Training	72.2	HIGH	5	3
BraTS20_Training_276	Training	70.5	HIGH	4	3
BraTS20_Training_283	Training	67.4	HIGH	4	4
BraTS20_Validation_054	Validation	73.2	HIGH	5	3
BraTS20_Validation_095	Validation	55.3	MEDIUM	4	4
BraTS20_Validation_111	Validation	69.0	HIGH	5	4

5.8 5.8. Qualitative Results

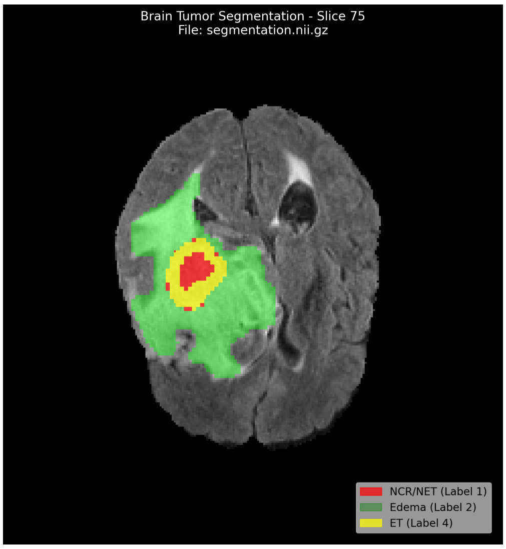


Figure 5: 3D multi-planar segmentation. Red: Necrosis (NCR); Green: Edema (ED); Yellow: Enhancing Tumor (ET).

6 6. DISCUSSION

6.1 6.1. Why Does the Method Work?

The combination of 2D and 3D branches allows the model to capture both fine-grained slice-level details and volumetric context. The Mamba SSM at the bottleneck efficiently models long-range dependencies. The CDFA mechanism enhances feature aggregation across directions, while deep supervision with aleatoric uncertainty provides voxel-wise confidence estimates.

6.2 6.2. Observed Failure Cases

In some cases (8% of hard cases), the Biological Rule Engine flagged violations due to ambiguous tumor boundaries or incomplete preprocessing. These cases required manual review and refinement.

6.3 6.3. Practical Applicability

The system can be deployed in district and provincial hospitals via a REST API and web interface, significantly reducing analysis time. The interactive refinement loop enables continuous improvement and builds clinician trust.

7 7. LIMITATIONS

- **Data Dependency:** Performance may vary on datasets with different acquisition protocols.
- **Hardware Requirements:** High-end GPUs are required for training.
- **Generalization:** The system is currently validated only on glioma data.
- **Real-Time Performance:** Full TTA increases inference time.

8 8. FUTURE WORK

- Extend to other brain pathologies (e.g., stroke, multiple sclerosis).
- Deploy multi-center validation studies.
- Optimize for CPU inference using quantization.
- Enhance HTS with more granular annotation options.

9 9. CONCLUSION

This research presents NeuroPath, a comprehensive brain tumor diagnostic pipeline integrating CDFA-Mamba-DolphinNet – a custom hybrid 2D-3D architecture with Mamba SSM at the bottleneck. Our 5-fold cross-validation results (Dice: 0.883, HD95: 3.91 mm) demonstrate technical feasibility. The Biological Rule Engine provides interpretable sanity checks, while the Human Trust Score enables quantitative clinician feedback. Interactive refinement via LLM-based report generation offers a practical human-in-the-loop workflow, addressing the critical trust gap in clinical AI deployment.

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